

SANDIP KARMAKAR

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📍 Konnagar, Hooghly, West Bengal, PIN Code - 712235, [IN]

PROFILE SUMMARY

PhD scholar in Electronics & Communication Engineering with expertise in IoT, wireless communication, and embedded systems. Experienced in healthcare monitoring solutions using Raspberry Pi and LoRa. Skilled in Python and machine learning, with research focus on 5G/6G networks and intelligent systems.

EXPERIENCE

- **Junior Research Fellow - (DETCE)** Sep 2025 - Ongoing
Jadavpur University 🌐 Jadavpur, Kolkata, WB - 700032, IN
 - Developing an IoT-based health monitoring prototype using Raspberry Pi, NodeMCU, Heltec LoRa, Python, Flask, and MongoDB for remote bio-signal acquisition and analysis.
- **Lecturer** Feb 2020 - May 2023
Salbani Institute of Technology 🌐 Salboni, Paschim Medinipur, WB - 721147, IN
 - Delivered lectures and lab sessions on Basic, Applied and Digital Electronics, Microprocessors and Microcontrollers; managed computer labs including OS/software installation, LAN setup, and network maintenance.

TECHNICAL SKILLS

- **Programming Languages:** Python, C Programming, JavaScript, Java, Verilog
- **Key Skills:** IoT, Machine Learning, Artificial Intelligence, SQL, Numpy, Pandas, Deep Learning, Image Processing, Data Visualization
- **Subjects:** Digital Logic, IoT, DBMS, Data Structure & Algorithm, VLSI, Microprocessor - Micro-controller, Cryptography, Embedded System
- **Software Skills:** VS Code, Jupyter Notebook, Eclipse, Omnet, Matlab, Xilinx, Proteus, Android Studio Code, Microsoft Office, Arduino IDE
- **Hardware Skills:** Raspberry Pi, Heltech WiFi LoRa, Computer Hardware and Networking, Arduino UNO

EDUCATION

- **PhD in Electronics & Communication Engineering** Jan 2026 - Ongoing
Jadavpur University 🌐 Jadavpur, Kolkata, WB - 700032, IN
 - Specialization: Communication
 - Topic: Studies of Massive MIMO and RIS-Assisted NOMA IoT Networks for 5G and Beyond Networks
 - GPA: NA
- **M. Tech in Distributed & Mobile Computing** Aug 2023 - June 2025
Jadavpur University 🌐 Jadavpur, Kolkata, WB - 700032, IN
 - GPA: 7.89/10.00
- **B. Tech in Electronics & Communication Engineering** Sep 2016 - Jul 2019
St. Thomas College of Engineering & Technology | MAKAUT 🌐 Khidirpur, Kolkata, WB - 700023, IN
 - GPA: 7.05/10.00
- **Diploma in Electronics & Telecommunication Engineering** Aug 2013 - Aug 2016
Technique Polytechnic Institute | WBCTVESD 🌐 Panchrokh, Sugandhya, Hooghly, WB - 712102, IN
 - GPA: 7.7/10.00

PROJECTS

- **Cost Effective Design and Implementation of Health Companion System for Human Bio-Signal Acquisition and Remote Monitoring in Rural/Urban Region: An IoT Prototype Model** Sep 2025 - Ongoing
Tools: Raspberry Pi 5, ESP32, Arduino UNO, Heltec WiFi LoRa 32, Sensors, Arduino IDE, C, Python, MongoDB, Flask Server
- **EMG Signal Acquisition and Transmission Over Low-Cost LoRa Wireless System: Analysis for Fatigue Muscle** Apr 2024 - May 2025
Tools: Arduino UNO, Heltec WiFi LoRa 32, Sensors, Arduino IDE, C, Python
- **Development of an IoT-Driven Wireless Smart Health Care System for Continuous Heart Monitoring** Apr 2024 - Feb 2025
Tools: Raspberry Pi 5, Heltec WiFi LoRa 32, Sensors, Arduino IDE, C, Python, MQTT Protocol
- **Automatic Unauthorized Parking Detector with SMS Notification** Jun 2018 - May 2019
Tools: Raspberry Pi, Pi Camera, Python Programming Language, Image Processing, SVM Classifier, Twilio Cloud
- **Design of Secured Wireless Real Time Electronic Voting Machine for Data Acquisition** Aug 2015 - May 2016
Tools: Microcontroller AT89C51, Keil C

PATENTS AND PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN SUBMISSION

- [C.1] Suman Paria, Sandip Karmakar, Iti Saha Misra. (2025). **Development of an IoT-Driven Wireless Smart Health Care System for Continuous Heart Monitoring.** In *Devices for Integrated Circuit (DevIC)*, Kalyani, India, 2025, pp. 727-730. DOI: 10.1109/DevIC63749.2025.11012139
- [J.2] Madhurima Chakrabarti, Sandip Karmakar, Sumit Singh, Madhura Sur, Dipankar Kundu. (May' 2019). **Automatic Unauthorized Parking Detector with SMS Notification.** *International Advanced Research Journal in Science, Engineering and Technology*, Vol. 6, Issue 5, pp. 152-157. DOI: 10.17148/IARJSET.2019.6522
- [J.1] Subhadeep Chakraborty, Sandip Karmakar, Rima jana , Subhradeep Dey. (May' 2019). **Design of Secured Wireless Real Time Electronic Voting Machine.** *INTERNATIONAL JOURNAL OF INNOVATIVE RESEARCH IN ELECTRICAL, ELECTRONICS, INSTRUMENTATION AND CONTROL ENGINEERING*, Vol. 3, Issue 9, pp. 27-32. DOI: 10.17148/IJIREICE.2015.3907

INDUSTRIAL TRAINING

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| • Introduction to IoT with AI-ML & Cloud Computing
<i>Indian Institute of Technology - Kharagpur Kharagpur, Paschim Medinipur, West Bengal</i> | <i>Feb 2025 - Aug 2025</i> |
| • Advance Data Science & Machine Learning
<i>Topstack Chandan Nagar, Hooghly, West Bengal</i> | <i>Jul 2023 - Feb 2024</i> |
| • FPGA based VLSI Design
<i>Sandeepani School of Embedded System Design Koramangala, Bangalore, Karnataka</i> | <i>Mar 2022 - Aug 2022</i> |
| • Advance Java and Java Spring Framework
<i>Topstack Chandan Nagar, Hooghly, West Bengal</i> | <i>Mar 2022 - Jul 2022</i> |
| • HTML, CSS, JAVA Script, Node JS, Mongo DB
<i>Topstack Chandan Nagar, Hooghly, West Bengal</i> | <i>Aug 2021 - Dec 2021</i> |
| • Industrial Electronics, PLC, Microcontroller
<i>G.E. Motors (Pvt.) Ltd. Sheoraphuli, West Bengal</i> | <i>7th Jun 2018 - 23rd Jun 2018</i> |
| • Basic Television Transmission System
<i>Praser Bharati, Doordarshan Kendra Kolkata, West Bengal</i> | <i>12th Oct 2015 - 16th Oct 2015</i> |

ADDITIONAL INFORMATION

Languages: Bengali (Native), English (Advanced), Hindi (Intermediate)

Soft Skills: Collaboration, Problem-solving, Time Management, Innovative, Team Lead, Efficient, Hard Working

Hobby: Painting, Swimming, Bike Riding